Hybrid Utility Function for Unexpected Recommendations

Pan Li
New York University
Stern School of Business
New York, NY, USA
pli2@stern.nyu.edu

ABSTRACT

Unexpectedness constitutes an important factor for recommender system to improve user satisfaction and avoid filter bubble issues. In this proposal, we propose to provide unexpected recommendations using the hybrid utility function as a mixture of estimated ratings, unexpectedness, relevance and annoyance. We plan to conduct extensive experiments to validate the superiority of the proposed method.

CCS CONCEPTS

• **Information systems → Recommender systems**.

KEYWORDS

Hybrid Utility, Unexpectedness, Recommender System

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1 MOTIVATION AND OVERVIEW OF RESEARCH DIRECTION

Exploring new perspectives of recommender system and analyzing its impact on consumers and business platforms have always been hot topics for recommender systems. Especially, the fields of unexpected recommendation [7, 9], multi-criteria recommendation [8, 14], cross-domain recommendation [6], sequence modeling [10, 16] and their applications have drawn lots of research attention.

In particular, this proposal focusing on the problem of **unexpected recommendations**, which serves as an important tool to break through filter bubbles and avoid the problem of user boredom, the goal of which is to provide novel yet satisfying recommendations to consumers. It aims to deviate from user expectations and manages to expand user interests. Providing unexpected recommendations has become a promising field in recommender systems, as researchers have introduced several evaluation metrics beyond accuracy, including unexpectedness, serendipity, novelty and diversity [13], the goal of which is to provide novel, surprising and not previously seen recommendations. Among them, unexpectedness is of particular interest for it is shown to be positively correlated with user satisfaction and helps to achieve strong recommendation performance [1]. Researchers propose to define unexpectedness of a recommended item as the distance between this item and the closure of previously consumed items, and manage to achieve great success along this direction.

However, previous unexpected recommendation models provide novel recommendations to users uniformly and globally, without considering the personality of different users. For example, some users might be ''variety-seekers" and tend to explore new items more than other users, who prefer to stay in their familiar fields. Therefore, a successful unexpected recommender system should take heterogeneous user preferences into consideration and intelligently select the optimal group of people to provide recommendations that deviate from their expectations.

In addition, it is also important to select the best occasion to provide unexpected recommendations instead of utilizing this strategy all the time. When the consumers are exposed to similar content over and over again in a short period, they might be eager to look for something different; meanwhile when they are comfortably enjoying their favorite movies, they do not wish to be interrupted by seemingly irrelevant recommendations. Therefore, we propose to use hybrid utility function to answer the following two research questions:

- **Which type of consumers need unexpectedness in recommendations?**
- **When do those consumers need unexpectedness in recommendations?**

2 BACKGROUND AND RELATED WORK

Unexpectedness measures the recommendations to users of those items that are not included in their consideration sets and depart from what they would expect from the recommender system. For example, it is defined as the distance between the recommended item and the set of expected items [1].

Researchers have shown the importance of incorporating unexpectedness in recommendations, which helps overcome the over-specialization problem [1, 5], broadens user preferences [4, 15] and increases user satisfaction [1, 11, 15]. They also propose effective unexpected recommendation models [3, 12] to capture the deviation of a recommender system from the results obtained from a primitive prediction model, including Serendipitous Personalized Ranking[11] that extends traditional personalized ranking methods by considering item popularity in AUC optimization; Auralist[15] that balances between the desired goals of accuracy, diversity, novelty and serendipity simultaneously; Determinantal Point Process (DPP) [2] that proposes a novel algorithm to greatly accelerate the greedy MAP inference and provide diversified recommendation

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accordingly; and HOM-LIN [1] that defines unexpectedness as the distance between items and the expected set of users.

However, these methods model unexpectedness in the feature space, while not considering the latent interaction between users and items as well as complex and heterogeneous relations from auxiliary information. Therefore in [7, 9], I propose to define unexpectedness as the distance metric in the latent space using latent feature and attribute embeddings. Specifically, the model takes the natural closure of convex hull in the latent space, and calculates unexpectedness as the euclidean distance from new item embedding to the latent convex hull of expected items of the user. The proposed unexpectedness measure is subsequently combined with estimated ratings for providing recommendations. It is capable of significantly improving novelty measures without losing any accuracy metric.

3 RESEARCH METHODOLOGY AND PROPOSED EXPERIMENTS

As discussed in the introduction section, it is important to target certain groups of people to provide unexpected recommendations, as opposed to uniformly suggest unexpected products to all users, as it might backfire for those who are more comfortable with their familiar items. Also, it is crucial to select the best occasions to guide the user to a different area. For example, it might not work if we disrupt the user with a set of seemingly unrelated video recommendations when she/he is enjoying the TV series. Therefore, to address these two research questions, I propose to incorporate the following components into the design of unexpected recommendations and the utility function for user i and item j:

$$Utility_{i,j} = EstRating_j + \alpha_i Unexp_{i,j} + Relevance_{i,j} + Annoyance_{i,j}$$

Sequence Behavior Modeling To capture the long-term user preferences based on historical behaviors, the recommendation model should take all the past transactions into account and utilize sequence modeling techniques to extract the latent essence of user preferences. We assume that long-term preference for each user is stable for each recommendation. Meanwhile, short-term session-based preference also plays an important role for the next-item recommendation problem. Thus, I propose to use a separate model to address short-term user intentions with specific focus on the interconnection between items within the session. Long-term and short-term user preferences are subsequently combined with representation of new items to estimate their ratings $EstRating_j$.

Personalized Unexpectedness As pointed out in [7, 9], it is beneficial to model unexpectedness in the latent space as opposed to the feature space. Therefore, we construct the expectation set for each user in the latent space to capture the complex relationships between them. Unexpectedness $Unexp_{i,j}$ is subsequently modeled as the distance between the new item and the expectation set. Note that each user has different preferences on unexpected recommendation results. Therefore, I propose to use the coefficient α_i to address the heterogeneity of users towards unexpectedness, which could be learned from past transactions to discover those ''variety-seekers".

Relevance It is worth noting that, on the one hand we wish to provide recommendations that users have not seen before; on the other hand, irrelevant recommendations might significantly reduce user satisfaction towards recommender system. Therefore, I propose to add relevance into the design of utility function as well to balance the effect of unexpectedness. Relevance $Relevance_{i,j}$ is defined as the distance between the new item and historic items in the latent space.

Annoyance Finally, I propose to specifically address the problem of boring and repeated recommendation using the term annoyance, which captures the similarity of new recommendations compared to the short-term user behavior in the same session. For example, if certain user has just bought a laptop online, and therefore receives recommendations of all types of laptops on the second day, those recommendations might really be annoying to that user.

To sum up, this proposal suggest to combine estimated ratings, personalized unexpectedness, relevance and annoyance to construct hybrid utility function and provide unexpected recommendations accordingly.

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